Structured questions

|!|BCELC0290000001|!|

The photograph below shows a Romer's Tree Frog (*Liuixalus romeri*), which is a native species in Hong Kong. It is the smallest amphibian among the frog and toad species recorded in Hong Kong.



(a) Define what a species is. (1 mark)

_____

_____

(b) The table below shows the taxonomic groups to which the Romer's Tree Frog belongs.

Taxonomic group	Name	Sequence
Kingdom		
Genus	*Liuixalus*	
Order	Anura	
Species	*Liuixalus romeri*	
Phylum	Chordata	
Family	Rhacophoridae	
Class	Amphibia	

(i) Write in the table the name of the kingdom to which Romer's Tree Frog belongs. (1 mark)

(ii) Write in column three the numbers 1 to 7 to show the sequence of taxonomic groups. Write the number 1 to show the group with the most organisms and the number 7 the least. (1 mark)

(c) Give ***two*** reasons why the Romer's Tree Frog is classified as an amphibian instead of a reptile. (2 marks)

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(d) The Romer's Tree Frog is predominantly nocturnal. State ***two*** advantages of being nocturnal. (2 marks)

(e) The population of the Romer's Tree Frog is decreasing due to human disturbance and habitat destruction. This may cause the loss of biodiversity. Why is it important to maintain biodiversity? (1 mark)

##

(a) A group of organisms that can interbreed to produce fertile offspring (1)

(b) (i) Animalia (1)

(ii) Starting from the top: 1, 6, 4, 7, 2, 5, 3 (1)

(c) Any ***two*** of the following: (2)

- The Romer's Tree Frog has moist skin without scales while reptiles have skin covered with hard, dry scales.
- The Romer's Tree Frog breathes with gills in its larval stage / breathes with lungs and skin in its adult stage while reptiles can only breathe with lungs.
- The Romer's Tree Frog carries out external fertilization while reptiles carry out internal fertilization.
- Reptiles lay eggs with a shell while the Romer's Tree Frog does not.

(Accept other reasonable answers.)

(d) Any ***two*** of the following: (2)

- To prevent dehydration
- To prevent over-heating
- To avoid predators

(Accept other reasonable answers.)

(e) Any ***one*** of the following: (1)

- To maintain the stability / ecological balance of ecosystems
- To maintain genetic variation within a species
- Humans depend on other organisms for resources
- Humans have an ethical responsibility to protect threatened species and prevent their extinction

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- A natural habitat with a high biodiversity has its aesthetic value

(Accept other reasonable answers.)

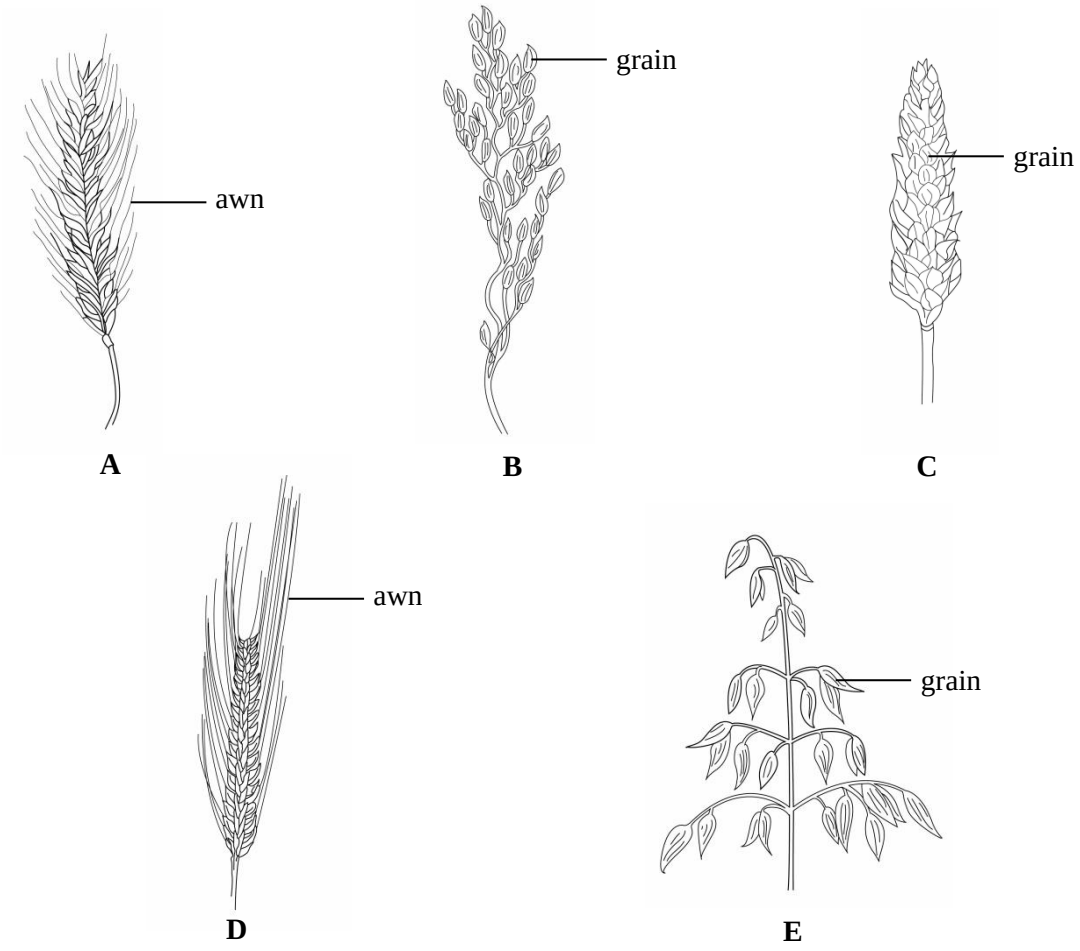
(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)	Understanding basic concepts	2
(d)	Applying concepts	2
(e)	Applying concepts	1

|!|BCELB0290000002|!| ★ **Cross topic**

The diagrams below show the fruiting heads of five types of cereals.



The key below is constructed for identifying the five types of cereals, A to E.

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1a awns present ..... go to 2
1b awns absent ..... go to 3
2a awns can be longer than the fruiting head ..... *Hordeum*
2b awns never longer than the fruiting head ..... *Secale*
3a grains closely packed in the fruiting head ..... *Triticum*
3b grains not closely packed in the fruiting head ..... go to 4
4a grains hanging down ..... *Avena*
4b grains not hanging down ..... *Oryza*

(a) Using the key, identify A, B and E. Write the sequence taken for making the identification. (3 marks)

Cereal	Name	Key sequence for identification
A		
B		
E		

(b) Most cereal plants are wind-pollinated. Suggest ***two*** ways in which cereal flowers may be different from insect-pollinated flowers. (2 marks)

(c) Describe the events that occur in flowering plants after fertilization to form a seed. (4 marks)

(d) What is the main food reserve stored in the grains of cereals? (1 mark)

##

(a) 1 mark for each row correct (3)

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Cereal	Name	Key sequence for identification
A	*Secale*	1a → 2b
B	*Oryza*	1b → 3b → 4b
E	*Avena*	1b → 3b → 4a

(b) Any ***two*** of the following: (2)

- Cereal flowers are smaller / less conspicuous / not as bright in colour as insect-pollinated flowers.
- Cereal flowers are unscented while insect-pollinated flowers are usually scented.
- Cereal flowers do not have nectaries to produce nectar while insect-pollinated flowers do.
- Cereal flowers have stigma and anthers hanging outside the flowers while insect-pollinated flowers have stigma and anthers enclosed within the flowers.
- Cereal flowers have feathery stigmas while insect-pollinated flowers have a broad sticky stigma.

(c) After fertilization, a zygote is formed. (1)
The zygote divides repeatedly by mitotic cell division to grow and develop into an embryo. (1)
The ovule develops into a seed, (1)
and the integuments become the seed coat. (1)

(d) Carbohydrates / Starch (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	2
(c)	Understanding basic concepts	4
(d)	Understanding basic concepts	1

|!|BCELC0290000003|!|

The photographs below show five different organisms.

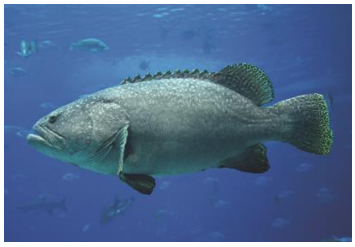
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A



B



C



D



E

(a) All the organisms shown above belong to the same domain and the same kingdom. State the domain and the kingdom. (2 marks)

(b) All the organisms shown above except A can be classified into one group. State the structural feature on which this classification is based. (1 mark)

(c) With reference to the organisms shown, name the respiration structure used by

(i) organism B only. (1 mark)

(ii) organism C only. (1 mark)

(d) Organisms D and E belongs to the same group. Name this group and give ***one*** distinctive feature of this group. (2 marks)

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(e) If organism D is stranded on a beach out of water, its respiratory structures cannot function properly. Explain why. (2 marks)

##

(a) Eukarya (1)
Animalia (1)
(b) Organism A does not have a backbone / vertebral column, while the others have. (1)
(c) (i) Skin (1)
(ii) Gills (1)
(d) Mammal (1)
They have mammary glands. (1)
(e) When organism D is out of water, its body weight is not supported by the buoyancy of water. (1)
The rib cage / lungs cannot expand. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	1
(c)(i)	Analysing data	1
(c)(ii)	Analysing data	1
(d)	Analysing data	1
	Understanding basic concepts	1
(e)	Applying concepts	2

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|!|BCELB0290000004|!| ★

The photographs below show six different animals.



A



B



C



D



E



F

(a) The animals shown belong to two different major groups.

(i) Name the two groups. (2 marks)

_____

_____

(ii) Classify animals A to F into the groups named in (i). (2 marks)

_____

_____

(b) Animals C and E belong to different sub-groups. Give ***two*** external features that are characteristic of the sub-group to which C belongs. (2 marks)

_____

_____

(c) The amount of daily food intake per unit mass of E is much greater than that of F. Explain this phenomenon. (4 marks)

_____

_____

_____

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##

(a) (i) Vertebrates, invertebrates (1, 1)

(ii) Vertebrates: C, E, F (1)

Invertebrates: A, B, D (1)

(b) It has feathers covering its body. (1)

It has a beak. (1)

(c) E is an endotherm (a homoiotherm) while F is an ectotherm (a poikilotherm). (1)

E has a higher body temperature than F, and so E loses more heat to the surroundings. (1)

E needs more food for respiration to release more energy (1)

to compensate for the heat loss. (1)

(Accept other reasonable answers.)

(10 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Analysing data	2
(b)	Analysing data	2
(c)	Applying concepts	4

|!|BCELB0290000005|!| ★ **Cross topic**

The table below shows some features of four specimens (A, B ,C and D) collected in a field study.

Features	A	B	C	D
Place where the specimen is found	on land	in water	in water	on decaying wood
Number of cells	multicellular	unicellular	unicellular	multicellular
Nucleus	✓	✓	✓	✓
Chlorophyll	✓	✓	×	×
Roots	✓	×	×	×
Stems	✓	×	×	×
Leaves	✓	×	×	×
Flowers	✓	×	×	×

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Hyphae	×	×	×	✓

Key: ✓ = present × = absent

(a) Why are all the four specimens classified in the same domain? (1 mark)

(b) Name the kingdom to which A, B, C and D belong respectively. (4 marks)

(c) Which of the specimens is probably an alga? (1 mark)

(d) (i) State and describe the mode of nutrition of D. (3 marks)

(ii) What is the ecological significance of D? (1 mark)

##

(a) They have a nucleus. (1)

(b) A: Plantae (1)
B: Protista (1)
C: Protista (1)
D: Fungi (1)

(c) B (1)

(d) (i) Saprophytic nutrition (1)
The hyphae of C secrete enzymes to break down the organic matter in the wood. (1)
Soluble nutrients are then absorbed into the cells. (1)

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(ii) C plays an important role in the recycling of materials in an ecosystem. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	4
(c)	Analysing data	1
(d)(i)	Understanding basic concepts	3
(d)(ii)	Understanding basic concepts	1

|!|BCELB0290000006|!| ★

The table below shows some features of four different vertebrates belonging to different groups.

	P	Q	R	S
Outer covering	wet and slimy scales	moist skin without scales	hair	feathers
Organs for gas exchange	gills	lungs	lungs	lungs
Structures for locomotion	fins	limbs	limbs	wings

(a) Using the information provided in the table, construct a dichotomous key for identifying the four vertebrates. (3 marks)

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(b) Which of P to S is a mammal? State ***one*** feature of mammals not listed in the table. (2 marks)

(c) Which of P to S is a bird? State ***one*** feature of birds not listed in the table. (2 marks)

(d) The photograph below shows a python.



(i) With reference to the photograph, give a reason why the python is not classified in the same group as Q. (1 mark)

(ii) Give ***two*** features that are common to the python and S. (2 marks)

##

(a) 1a have lungs for gas exchange ..... go to 2
1b have gills for gas exchange ..... P
2a with hair or feathers ..... go to 3
2b without hair or feathers ..... Q
3a body covered with hair ..... R
3b body covered with feathers ..... S
(1 mark for each pair of statements) (3)

(b) R (1)
Mammals have mammary glands. (1)

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(c) S (1)

Birds have a beak. (1)

(d) (i) The python has hard, dry scales covering its body. (1)

(ii) Any ***two*** of the following: (2)

- Both have lungs for gas exchange.
- Both carry out internal fertilization.
- Both lay eggs with hard shells.

(10 marks)

##

	Skills assessed	Marks
(a)	Presenting information	3
(b)	Analysing data	1
	Understanding basic concepts	1
(c)	Analysing data	1
	Understanding basic concepts	1
(d)(i)	Analysing data	1
(d)(ii)	Understanding basic concepts	2

|!|BCELA0290000007|!| ★★ **Cross topic**

The photograph below shows a chicken.



(a) Explain why chickens are classified in the kingdom Animalia. (2 marks)

(b) The chicken has feathers covering its body. What is the function of feathers to the chicken? (1 mark)

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(c) What type of fertilization do chickens have? (1 mark)

(d) Feather colour in chickens is controlled by a gene located on an autosome. The gene has two alleles which are codominant. The allele C^B in the homozygous condition produces black feathers. The allele C^W in the homozygous condition produces white feathers.

A cross between a chicken with black feathers and a chicken with white feathers produced F_1 offspring all with grey feathers.

(i) What is meant by codominant alleles? (1 mark)

(ii) The F_1 offspring were allowed to interbreed. Draw a genetic diagram to show the possible genotypes and phenotypes of the F_2 offspring. (3 marks)

(iii) Are the chicken with black feathers and the chicken with white feathers in the parental generation regarded as the same species? Give a reason to support your answer. (1 mark)

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##

(a) Any ***two*** of the following: (2)

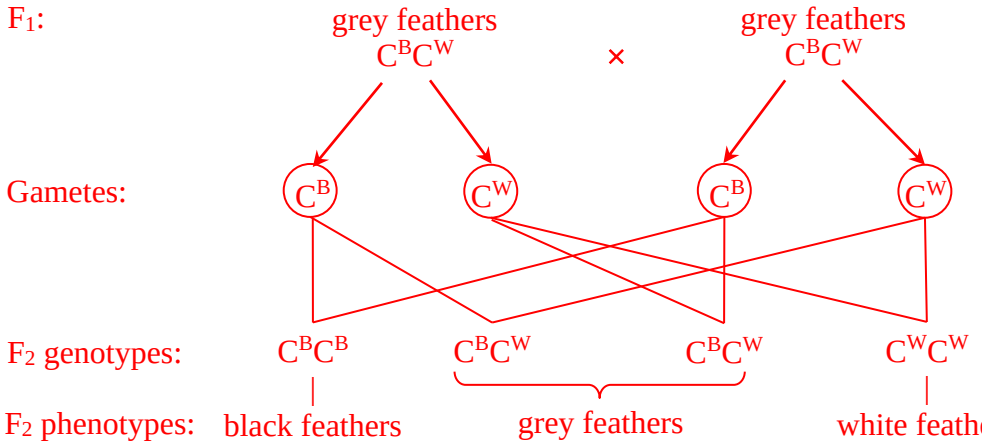
- They are multicellular.
- Their cells do not have a cell wall.
- They are heterotrophic / feed on other organisms.

(b) Feathers trap air which helps reduce heat loss in cold weather. (1)

(c) Internal fertilization (1)

(d) (i) Both alleles are expressed in a heterozygote. (1)

(ii)



(1)

(1)

(1)

(iii) Yes

Because they can interbreed and produce fertile offspring (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)(i)	Understanding basic concepts	1
(d)(ii)	Presenting information	3
(d)(iii)	Applying concepts	1

|!|BCELC0290000008|!|

The dichotomous key below can be used to identify five types of plants.

1a with vascular tissues ..... go to 2

1b without vascular tissues ..... A

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2a	produce seeds	go to 3
2b	do not produce seeds	B
3a	produce flowers	go to 4
3b	do not produce flowers	C
4a	with one cotyledon in the seeds	D
4b	with two cotyledons in the seeds	E

(a) Which of the above plants is restricted to grow in moist places? Explain why. (3 marks)

(b) Name ***one*** structure that is found in C but not in the other four types of plants. State the function of this structure. (2 marks)

(c) Give ***two*** differences between plants B and D. (2 marks)

(d) Apart from the number of cotyledons, give ***two*** differences between plants D and E. (2 marks)

##

(a) A (1)

It lacks vascular tissues for transportation of water throughout the plant. (1)

It is restricted to grow in moist places so that water is directly absorbed from the plant surfaces. (1)

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(b) Cone (1)

It carries seeds for reproduction. (1)

(c) Any ***two*** of the following comparisons: (2)

Plant B	Plant D
• reproduced by spores	• reproduced by seeds
• spores enclosed in capsules	• seeds enclosed in fruits
• smaller	• larger
• anchored by rhizoids	• anchored by roots
• no vascular tissues / xylem / phloem	• has vascular tissues / xylem / phloem

(Accept other reasonable answers.)

(d) Any ***two*** of the following comparisons: (2)

Plant D	Plant E
• parallel venation	• net venation
• fibrous roots	• taproots
• number of petals usually in multiples of three	• number of petals usually in multiples of four or five

(Accept other reasonable answers.)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Analysing data	1
	Understanding basic concepts	1
(c)	Analysing data	2
(d)	Analysing data	2

|!|BCELB0290000009|!| ★

Figure 1 shows a mayfly.

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Figure 1



[**Photo source:** Hectonichus / CC-BY-SA-3.0]

The classification hierarchy for this mayfly from top to bottom is:

Animalia
Arthropoda
Insecta
Ephemeroptera
Ephemeridae
Ephemera danica

(a) State the genus of this mayfly. (1 mark)

_____

(b) The mayfly does not have a vertebral column to support its body. What type of support system does it has? (1 mark)

_____

(c) The mayfly belongs to the order Insecta. Give ***two*** common features of organisms in this order. (2 marks)

_____

_____

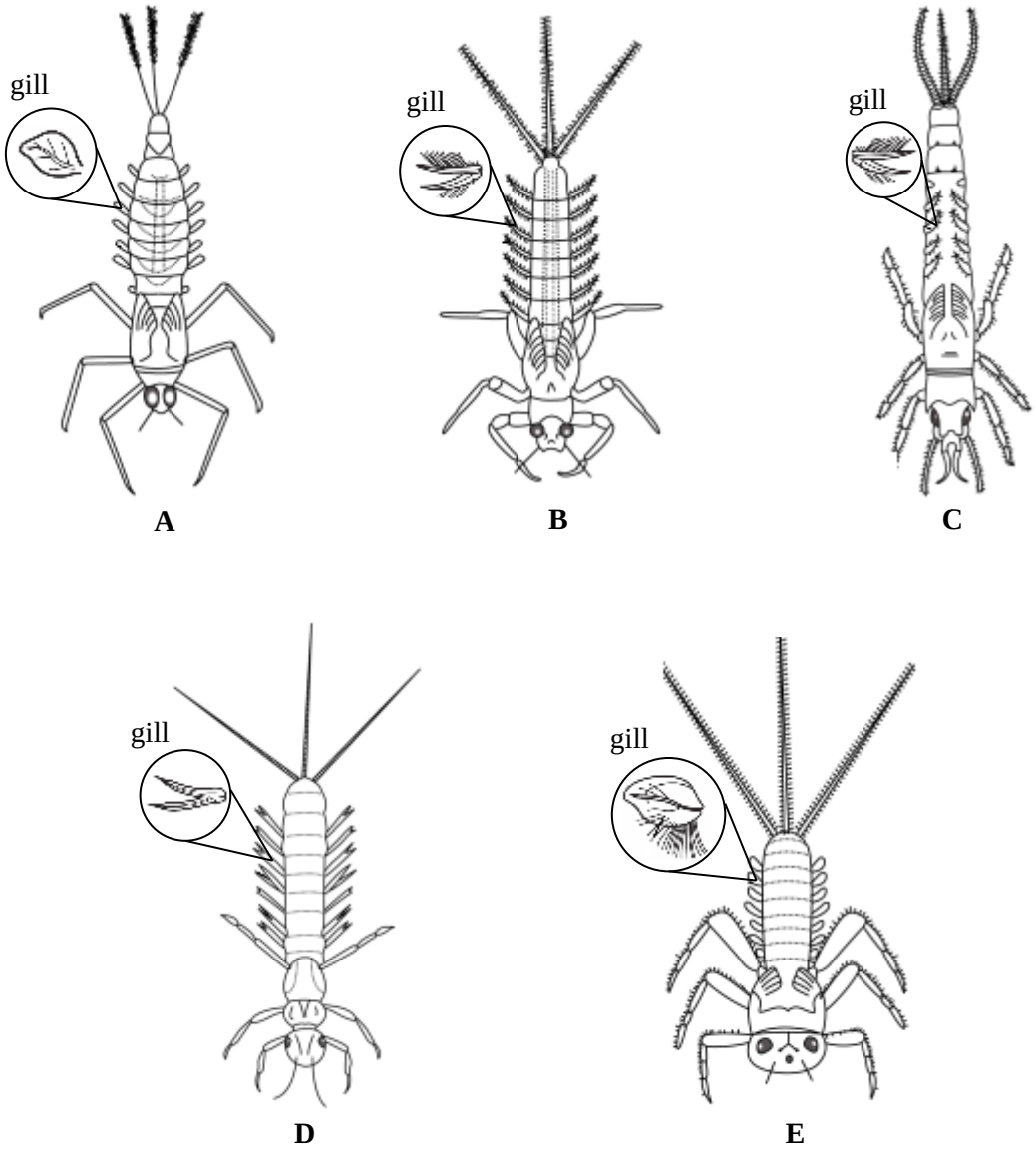
_____

_____

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(d) Mayflies lay their eggs in water. The eggs hatch into larvae called nymphs. Figure 2 shows the nymphs of five types of mayflies.

Figure 2



(i) Apart from having gills for gas exchange, suggest another adaptive feature shown by the nymphs in Figure 2 that enables them to live in water. (1 mark)

_____

_____

(ii) Use the dichotomous key to identify the nymphs A to E by genera. Complete the table below by matching the nymphs to their genera. (3 marks)

1a Rear pair of legs pointing towards tails ..... go to 2
1b Rear pair of legs pointing forwards or sideways ... go to 3

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2a Gills projecting sideways from body ..... *Paraleptophlebia*
2b Gills folded over body ..... *Ephemera*
3a Each gill being a single flat plate ..... go to 4
3b Each gill divided into two strands ..... *Potomanthus*
4a Tails feather-like in shape ..... *Centroptilum*
4b Tails needle-shaped ..... *Ecdyonurus*

Genus	Nymph
Paraleptophlebia	
Ephemera	
Potomanthus	
Centroptilum	
Ecdyonurus	

##

(a) *Ephemera* (1)

(b) Exoskeleton (1)

(c) Their bodies are divided into three parts: head, thorax and abdomen. (1)
They have three pairs of jointed legs. (1)
(Accept other reasonable answers.)

(d) (i) Legs / Tails for propulsion (1)

(ii)

Genus	Nymph
Paraleptophlebia	D
Ephemera	C
Potomanthus	B
Centroptilum	A
Ecdyonurus	E

(Deduct 1 mark for every wrong answer.) (3)

(8 marks)

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Understanding basic concepts	2
(d)(i)	Analysing data	1
(d)(ii)	Analysing data	3

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In a field trip, students made observations of five organisms, P to T. They reported their observations as follows.

Student A: I found the organism (organism P) in water. It swims with its fins.

Student B: The organism (organism Q) stood on the ground when I saw it. It walked on two legs which were covered by dry scales. It flew away when I walked nearby.

Student C: I found the organism (organism R) was laying eggs under a stone. It has dry, hard scales covering its skin. The colour of its skin changed from green to brown.

Student D: The organism (organism S) appeared as some green patches on the water surface of a pond. I could not see it has any stem or leaf structures.

Student E: The organism (organism T) was brownish black in colour. It had a long and segmented body with two pairs of walking legs on each segment. It had more than ten segments.

(a) Based on the above observations, deduce which of the organisms

(i) is/are vertebrate(s). (1 mark)

(ii) is/are endothermic. (1 mark)

(iii) use(s) gills for gas exchange. (1 mark)

(b) Explain the significance of organism R being able to change the colour of its skin. (2 marks)

(c) Student D thinks that organism S is a plant. Is the student correct? Explain your answer. (2 marks)

(d) Student E thinks that organism T is an insect. Is the student correct? Explain your answer. (2 marks)

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##

(a) (i) P, Q and R (1)

(ii) Q (1)

(ii) P (1)

(b) Any ***one set*** of the following: (2)

- This is used as camouflage to blend in with the environment (1)
  so that it can hide from predators / prey. (1)

OR

- This is used for temperature regulation. (1)
  This skin may change to a darker color to absorb light and heat to raise their temperature / to a lighter color to reflect light and heat. (1)

(Accept other reasonable answers.)

(c) No, (1)

because organism S has no stem or leaf structures. (1)

(d) No, (1)

because organism T has more than three body segments / three pairs of legs. (1)

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(a)(iii)	Analysing data	1
(b)	Applying concepts	2
(c)	Analysing data	2
(d)	Analysing data	2

|!|BCELB0290000011|!| ★

In the 18th century, the Swedish biologist Carolus Linnaeus established a hierarchical classification system and the binomial system of nomenclature for plants and animals.

(a) What is Linnaeus' classification system based on? (1 mark)

(b) According to Linnaeus' classification system, what is meant by a species? (1 mark)

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(c) Give ***one*** example of organism to which the meaning of species in (b) is ***not*** applicable, and explain why. (2 marks)

(d) State what is meant by the binomial system of nomenclature, and explain why this is important. (3 marks)

(e) Modern classification is based on the phylogenetic relationships of organisms. Explain how DNA sequencing can provide evidence for determining the phylogenetic relationships of organisms. (2 marks)

##

(a) Structural / Physiological features of organisms (1)

(b) A species is a group of organisms which can interbreed and produce fertile offspring. (1)

(c) Bacteria / Archaebacteria (or other reasonable answers) (1)
These organisms reproduce by asexual binary fission / asexual reproduction and do not interbreed. (1)

(d) Each type of organism is given a scientific name, (1)
consisting of a genus name and a species name. (1)
This allows the establishment of an internationally recognized naming system. / This allows scientists to communicate accurately. (1)

(e) DNA sequencing determines the nucleotide sequence of DNA of different organisms. (1)
It is assumed that organisms that share more similarities in DNA sequence have a closer phylogenetic relationship. (1)

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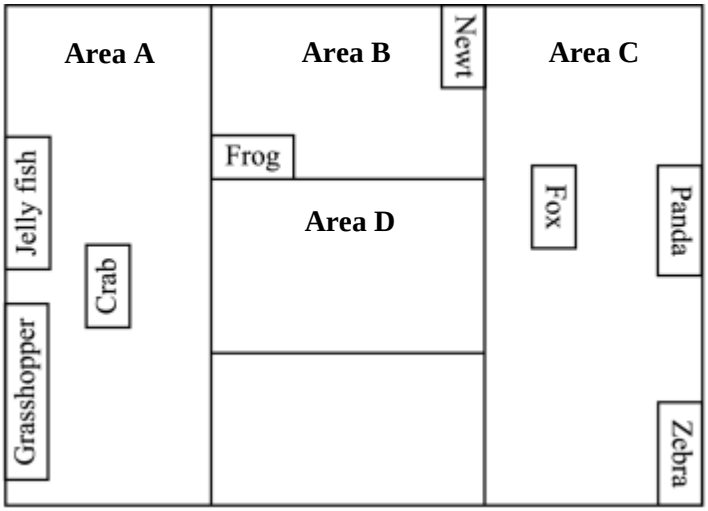
(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Applying concepts	2
(d)	Understanding basic concepts	3
(e)	Applying concepts	2

|!|BCELB0290000012|!| ★ **Cross topic**

The diagram below shows the floor plan of a natural science museum. The animals are kept in different exhibition areas according to their animal group.



(a) Give ***one*** feature of the animals displayed in area A that distinguishes them from animals in other exhibition areas. (1 mark)

(b) Explain how the animals displayed in area B are adapted to live both on land and in water. (2 marks)

(c) Give ***two*** common structural features which are characteristic of the animals displayed in area C. (2 marks)

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(d) Only a small number of pandas exist in the wild today and it is listed as a protected species. Suggest ***one*** conservation measure to maintain or increase its population. (1 mark)

(e) Area D is reserved for displaying animal X shown in the photograph below.



(i) Name the vertebrate group to which X belongs. (1 mark)

(ii) Frogs and X are adapted to live both on land and in water. Explain why frogs must return to the water to reproduce whereas X does not. (2 marks)

##

(a) They do not have a backbone / vertebral column. (1)

(b) They breathe with lungs on land (1)
and use the surface of the skin to carrying out gas exchange in water. (1)

(c) They have mammary glands / hair covering their bodies / pinnae. (any ***two***) (2)

(d) Setting up a natural reserve / Captive breeding / Legislation (or other reasonable answers) (1)

(e) (i) Reptiles (1)

(ii) Any ***two*** of the following: (2)

- Frogs carry out external fertilization while X carries out internal fertilization.

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- Frogs need water to provide a medium for the male gametes (sperms) to reach the female gametes (eggs) while X does not.
- Fertilized eggs of frogs would dry out on land while the fertilized eggs of X have shells to prevent desiccation.

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	2
(c)	Analysing data	2
(d)	Applying concepts	1
(e)(i)	Analysing data	1
(e)(ii)	Applying concepts	2

|!|BCELB0290000013|!| ★

The photographs below show the leaves of four different plants, A to D.



A



B



C



D

[**Photo source:** Forest and Kim Starr /CC-BY-3.0-US]

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(a) Based on the external features of the leaves of the above plants, construct a dichotomous key for identification of the plants. (3 marks)

(b) The plants shown in the photographs belong to two different plant groups.

(i) Name the two groups. (2 marks)

(ii) Classify plants A to D into the groups named in (i). (2 marks)

(iii) Give ***one*** similarity and ***one*** difference between the plant groups. (2 marks)

##

(a) 1a Leaves are needle-shaped ..... A
1b Leaves are not needle-shaped ..... go to 2
2a Parallel veins in leaves ..... B
2b Branched veins in leaves ..... go to 3
3a Leaves with lobes C

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3b Leaves without lobes ..... D

(1 mark for each pair of statements) (3)

(b) (i) Conifers, flowering plants (1, 1)

(ii) Conifer: A (1)

Flowering plants: B, C, D (1)

(iii) Similarity (any ***one***): (1)

- Both have vascular bundles.
- Both have true roots, stems and leaves.
- Both have seeds.

Difference (any ***one***): (1)

- Flowering plants produce flowers for reproduction while conifers do not.
- The seeds of flowering plants are protected inside fruits while the seeds of conifers are naked / carried by cones.

(9 marks)

##

	Skills assessed	Marks
(a)	Presenting information	3
(b)(i)	Analysing data	2
(b)(ii)	Analysing data	2
(b)(iii)	Understanding basic concepts	2

|!|BCELB0290000014|!| ★ **Cross topic**

The photographs below show five types of organisms in the South Pole region.

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A



B

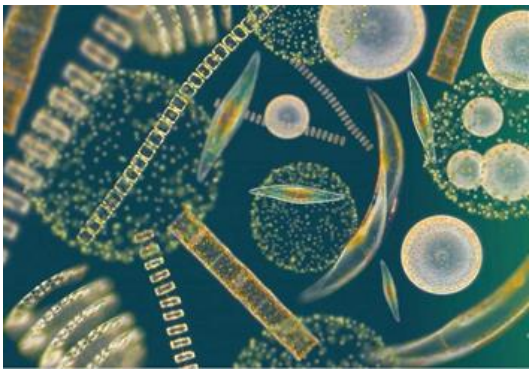
[Photo source: Uwe Kils /CC-BY-SA-3.0]



C



D



E

(a) All the organisms shown except E belong to the same kingdom.

(i) Name this kingdom. (1 mark)

_____

(ii) State ***two*** features common to all members in the kingdom in (i). (2 marks)

_____

_____

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(b) Both organisms A and B are filter-feeders. Why do they belong to different major groups? (1 mark)

(c) Two of the organisms shown belong to the same groups as humans.

(i) Identify the two organisms. (1 mark)

(ii) State ***one*** feature that identifies an organism as a member in this group. (1 mark)

(d) Explain ***one*** physiological adaptation and ***one*** structural adaptation of organism C to survive the cold climate. (4 marks)

(e) In the South Pole region, the period of daylight becomes much shorter in winter. Explain how this affects the population of E. (2 marks)

##

(a) (i) Animalia (1)

(ii) Any ***two*** of the following: (2)

- They are multicellular.
- Their cells do not have a cell wall.
- They are heterotrophic / feed on other organisms.

(b) A has a backbone / vertebral column while B does not. (1)

(c) (i) A and D (1)

(ii) Possession of mammary glands / hairs (1)

(d) Physiological adaptation:

- It is endothermic. (1)

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It can generate body heat internally by metabolism/respiration to maintain a relatively constant body temperature. (1)

Structural adaptation (any ***one set*** of the following): (2)

- It has feathers covering its body.
  Feathers trap air which helps reduce heat loss from the body.

OR

- It has a thick layer of subcutaneous fat
  which helps reduce heat loss from the body / acts as an energy reserve.

(e) The shorter period of daylight would result in a lower photosynthetic activity. (1)

Hence, the population of E would decrease. (1)

(12 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	2
(b)	Analysing data	1
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	1
(d)	Applying concepts	4
(e)	Applying concepts	2

|!|BCELB0290000015|!| ★ **Cross topic**

The table below shows the classification of three animals which belong to the class Mammalia. The classification levels are in hierarchical order.

	Chinese white dolphin (Indo-Pacific humpback dolphin) 	**Humpback whale** 	**Killer whale** 
i	Artiodactyla	Artiodactyla	Artiodactyla
ii	Delphinidae	Balaenopteridae	Delphinidae
Genus	*Sousa*	*Megaptera*	*Orcinus*
Species	*chinensis*	*novaeangliae*	*orca*

(a) Name the classification levels **i** and **ii** shown in the table. (2 marks)

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(b) Each organism is given a scientific name according to the binomial system of nomenclature.

(i) Give the scientific name of the Chinese white dolphin. (1 mark)

(ii) Explain the advantage of using scientific names over common names. (2 marks)

(c) Which two of the above animals have the closest phylogenetic relationship? Explain your answer. (2 marks)

(d) The photograph below show a carp.



According to modern classification systems, dolphins and carps belong to different vertebrate groups. State ***two*** structural differences between these two animals that justify this classification. (2 marks)

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(e) Only a small number of Chinese white dolphins exist today and it is listed as a protected species in Hong Kong. Suggest ***one*** human activity that may pose threats to its survival in Hong Kong waters. (1 mark)

_____

_____

##

(a) **i**: Order (1)

ii: Family (1)

(b) (i) *Sousa chinensis* (1)

(ii) Each species has only one scientific name which is universal. (1)

Using scientific names can prevent confusion when scientists from all over the world communicate with each other. (1)

(c) The Chinese white dolphin and the killer whale (1)

They belong to the same family while the other animal shares the same classifications levels at higher levels with them. (1)

(d) Any ***two*** of the following differences: (2)

Dolphin	Carp
• presence of mammary glands	• absence of mammary glands
• presence of lungs (absence of gills)	• absence of lungs (presence of gills)
• body not covered by scales	• body covered by scales

(e) Any ***one*** of the following: (1)

- Reclamation in the coastal areas / Construction / Dredging of seabed causes deterioration of the water quality of their natural habitats.
- Disturbance caused by tourism drives the dolphins away from their natural habitats.
- Fishing activity may accidentally kill or injure the dolphins.
- Heavy sea traffic may drive away the dolphins / accidentally injure the dolphins / interfere with their food-finding.

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	2
(c)	Analysing data	2
(d)	Understanding basic concepts	2
(e)	Applying concepts	1

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|!|BCELC0290000016|!|

The photographs below show five different kinds of organisms.



A



B



C



D



E

(a) (i) Why are organisms A and B a classified in the same kingdom? (2 marks)

(ii) Compare the mode of nutrition and location of organisms A and B. (4 marks)

	A	**B**
Mode of nutrition		
Mode of locomotion		

(b) C is a type of bacteria commonly found in the human intestine. D lives in hot sulphur springs. They are both prokaryotes but belong to different kingdoms.

(i) Name the kingdom to which D belongs. (1 mark)

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(ii) Give ***one*** structural difference between C and D. (1 mark)

_____

_____

(c) Organism E was once classified in the kingdom Plantae. Based on some recent findings, it was classified in a separate kingdom.

(i) Suggest ***one*** reason why E was classified in the kingdom Plantae. (1 mark)

_____

_____

(ii) Suggest ***one*** reason why E was re-classified. (1 mark)

_____

_____

##

(a) (i) Both are unicellular (1)
and have a nucleus / have membrane-bounded organelles / are eukaryotic. (1)

(ii)

	A	B
Mode of nutrition	heterotrophic	autotrophic / phototrophic / heterotrophic
Mode of locomotion	pseudopods / amoeboid movement	flagella

(4)

(b) (i) Archaebacteria (1)

(ii) Any ***one*** of the following: (1)

- The cell wall of C is made up of peptidoglycan while the cell wall of D does not contain peptidoglycan.
- The lipids that make up the cell membrane of C and D are different.

(Accept other reasonable answer.)

(c) (i) It cannot move / is sessile like plants. (1)

(ii) Any ***one*** of the following: (1)

- The cell wall of plants is made up of cellulose while the cell wall of E is made up of chitin.
- Plants have chlorophyll while E does not have chlorophyll.

(Accept other reasonable answer.)

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(10 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Analysing data	4
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)(i)	Analysing data	1
(c)(ii)	Analysing data	1

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